Algorithm:
1 B
1. stort the program.
2. Function Definition: is - Poume (num)
Input: A number num.
check for primality:
a. if num is less than 2:
Return False (the number is not prime)
b. for each integer 1 from 2 to the
square root of num Cindusive).
if num is divisible by ici.e., num%i==0)
Return false (the number is not prime).
c. if no divisore core found:
Return Trie (the number is poume.
3. Main Program:
Through a. Loop through Numbers:
for each number/num in the range from
2 to 19
4. Call the function is. Prime (num).
(Inclusive):
a. if the fimction retwone wave (the
number is
b. Print the number.
Primes:
5. Stop the Program.
How chant
Start
Ifnum 2. For i in range 2,int (num=0,+1
fom num in range (2.20)
If is Prime (num)
Print (num)
End
PROGRAM:

```
def is_prime(num):
if num < 2:
return False
for i in range(2, int(num **0.5) + 1):
if num % i ==0:
return False
return True
for num in range(2, 20):
if is_prime(num):
print(nur)
```

Output:2.3.5.7.11.13.17.19
Result
The above program has executed and
verified successfully

Algorithm

2 B

1.start the program

2.read a value number to find factorial

3.invoke the function factorial to find the factorial of the number

A.if x==,then return 1

B.otheruise return x factorial(x-1)this function recursively invoke itself until x==0

4.displays the factorial of the given number

5.stop the program

Flow chart

Start

Read n

i=1

Fact-1

ifn==0 return n*factorial (n-1)

return 1

Stop

PROGRAM:

```
def Factorial(n):
if n==0:
return 1
else:
return n*Factorial(n-1)
print(format("FACTORIAL-RECURSIVE",
'^40'))
n= int(input("Enter number to find
factorial:"))
print("Factorial of",n,"is:", Faetorial(n))
```

OUTPUT:

FACTORIAL-RECURSIVE

Enter number to find factorial:5

Factorial of 5 is: 120

Result:

The above program has executed and verified successfully

Algorithm:

1. Start the Program

2. Initialize 1=1.

3. Iterate the outer Loop while for i<=٩٠

a. Initialize j=1.

b. Iterate the inner Loop while for j<=i

i. Display the value of i in the same line.

C. Go to the new line.

4. Stop the program.

Flow chart

Start

Input n<10

Isn< 10

Loop 1 = o to n

String = " "

Loop j=1 to 1-1

string + = "#"

Print string centered

Loop i = n+1 to 1 (reverce)

String = "

Loop j=1 to 1-1

string

Print string (centres)

End

Program

```
n=int(input("Enter a value less than 10:"))
for i in range (0,n+1):
string=""
for j in range(1,i):
string=string+"*"
print (format(string,"^10"))
for i in range(n+1,0,-1):
string=""
for j in range(1,1):
string=string+"*"
print(format(string,"^10"))
```

Output. enter a less than 10:8

*1 to 8 to1

Result

The above program has executed and verified successfully

4 B